



Nan Rosenbloom <nanr@ucar.edu>

No Krakatoa Runs

27 messages

Gokhan Danabasoglu <gokhan@ucar.edu>

Wed, Mar 1, 2023 at 7:09 AM

To: Nan Rosenbloom <nanr@ucar.edu>, Mike Mills <mmills@ucar.edu>

Cc: Gokhan Danabasoglu <gokhan@ucar.edu>, Isla Simpson <islas@ucar.edu>

Nan and Mike,

To further understand the memory associated with ocean initial conditions in the CESM2 Large Ensemble set, after another set of discussions, we think that having an ensemble of simulations without the Krakatoa eruption would be very useful. Presumably, these will be very similar to the no-Pinatubo simulations. The idea would be to start from the nearest restart files before the eruption and run them for 10-15 years or so. Ideally, we should do this for all four micro start years and 10 ensembles each. We can use the CESM Emerging Science allocation for this purpose. Happy to chat further.

Isla: Please chime in with any further details.

Best,
Gokhan

Nan Rosenbloom <nanr@ucar.edu>

Wed, Mar 1, 2023 at 7:51 AM

To: Gokhan Danabasoglu <gokhan@ucar.edu>

Cc: Mike Mills <mmills@ucar.edu>, Gokhan Danabasoglu <gokhan@ucar.edu>, Isla Simpson <islas@ucar.edu>

Mike ~

I am guessing that you are **quite busy**; let me know how I can help (if at all!). :)

Nan

~Nan Rosenbloom

nanr@ucar.edu

ORCID ID =0000-0001-7389-3346

She/her/hers ([Why do I share this info?](#))

[Quoted text hidden]

Mike Mills <mmills@ucar.edu>

Wed, Mar 1, 2023 at 12:37 PM

To: Gokhan Danabasoglu <gokhan@ucar.edu>

Cc: Nan Rosenbloom <nanr@ucar.edu>, Isla Simpson <islas@ucar.edu>

Hi all,

There are two ways to approach these experiments.

1. Re-run WACCM without Krakatoa to generate the forcings for CESM2.

This would be the most rigorous and smooth approach. It would just cost a bit a computing time.

The historical forcings for CESM2 were created from the average of the 3 WACCM historical runs to smooth out modes of variability. So the most rigorous approach would be to run 3 WACCM runs without Krakatoa for 10 to 15 years each post-Krakatoa. However, a single run might be good enough if computing resources are short. And the run(s) could be for a shorter period (maybe 5 years each post-Krakatoa?) and patched in to the existing historical forcings.

2. Patch the 1850 piControl forcings in to the existing historical forcings.

This would not require any new WACCM runs. However, there was a significant eruption just 3 years after Krakatoa (1886), so that would involve some decisions.

We could switch back to the existing historical forcings at the time of that eruption, which would then include the tail end of Krakatoa's impacts. These may or may not be very significant 3 years after the eruption. I could look at AOD plots to give some indication.

Alternately, we could decide to erase the 1886 eruption as well, and use the piControl forcings for a longer period.

Also, there might be further implications of switching back to 1850 forcings for in 1886, although this would not affect greenhouse gases. The affected forcings are stratospheric ozone, stratospheric aerosols, tropospheric oxidants, stratospheric water production from methane oxidation, and nitrogen deposition of land and ocean models.

Let me know what your thoughts are. I'm happy to talk about this in a meeting as well.

Mike Mills (he/him/his)

CESM WACCM Community Liaison

Atmospheric Chemistry Observations & Modeling Laboratory

National Center for Atmospheric Research

Phone: 303-497-1425

<https://acomstaff.acon.ucar.edu/mmills/>CESM forum: <https://bb.cgd.ucar.edu/cesm/>

[Quoted text hidden]

Isla Simpson <islas@ucar.edu>

Wed, Mar 1, 2023 at 3:09 PM

To: Mike Mills <mmills@ucar.edu>

Cc: Gokhan Danabasoglu <gokhan@ucar.edu>, Nan Rosenbloom <nanr@ucar.edu>

Hi Mike,

Thanks. Do you know which of these were used for the no pinatubo experiments? While I don't think it matters for our particular purpose, it might be nice to have a consistent methodology between the no Pinatubo and the no Krakatoa experiments. Or maybe we should at least take that into consideration. Gokhan, do you think it may also be worth consulting with Clara and Who as to whether they think there's an important role for the 1886 eruption in what they are seeing? I don't have a good sense of what we should do there given what we're looking at but my feeling is we maybe want to omit it too but I think it could be worth confirming that with them.

Thanks,
Isla

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Isla Simpson
Scientist 3, Climate Analysis Section
Climate and Global Dynamics Laboratory
National Center for Atmospheric Research
P.O. Box 3000 Boulder, CO 80307
Tel: 303-497-1763
<http://www.cgd.ucar.edu/staff/islas/>

Gokhan Danabasoglu <gokhan@ucar.edu>

Wed, Mar 1, 2023 at 3:30 PM

To: Isla Simpson <islas@ucar.edu>

Cc: Mike Mills <mmills@ucar.edu>, Nan Rosenbloom <nanr@ucar.edu>, Gokhan Danabasoglu <gokhan@ucar.edu>

Dear All,

For Pinatubo, we took the easy route: Mike's second option. Whatever is the easiest, we should follow that.

I will check with Clara and Who about the 1886 eruption.

Best,
Gokhan

[Quoted text hidden]

Mike Mills <mmills@ucar.edu>

Wed, Mar 1, 2023 at 5:55 PM

To: Gokhan Danabasoglu <gokhan@ucar.edu>

Cc: Isla Simpson <islas@ucar.edu>, Nan Rosenbloom <nanr@ucar.edu>

Thanks, Gokhan. I'm not recalling whether I was involved with the no Pinatubo runs. If the forcings switched from 1990 to 1850 levels, there might be some discontinuities, such as a drop in the non-volcanic stratospheric aerosols. But these aerosol levels would be very low anyway.

I assume then that only the stratospheric aerosols were changed to remove Pinatubo, and not stratospheric ozone, tropospheric oxidants, or other forcings. Is there a namelist I could look at for those runs?

Mike

Mike Mills (he/him/his)

CESM WACCM Community Liaison

Atmospheric Chemistry Observations & Modeling Laboratory

National Center for Atmospheric Research

Phone: 303-497-1425

<https://acomstaff.acom.ucar.edu/mmills/>

CESM forum: <https://bb.cgd.ucar.edu/cesm/>

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Gokhan Danabasoglu <gokhan@ucar.edu>

Wed, Mar 1, 2023 at 6:48 PM

To: Mike Mills <mmills@ucar.edu>

Cc: Isla Simpson <islas@ucar.edu>, Nan Rosenbloom <nanr@ucar.edu>, Gokhan Danabasoglu <gokhan@ucar.edu>

Mike,

I think Nan has a record of what exactly was done with the no Pinatubo runs.

Best,
Gokhan

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Nan Rosenbloom <nanr@ucar.edu>

Wed, Mar 1, 2023 at 9:00 PM

To: Gokhan Danabasoglu <gokhan@ucar.edu>

Cc: Mike Mills <mmills@ucar.edu>, Isla Simpson <islas@ucar.edu>, Gokhan Danabasoglu <gokhan@ucar.edu>

Hi Mike -

Here is the CASEROOT for one of the noPinatubo simulations, as well as the namelist.

CASEROOT:

/glade/work/nanr/CESM2-noPinatubo/b.e21.BHISTsmbb.f09_g17.LE2-1251.011-noPinatubo.001

user_nl_cam:

prescribed_strataero_datapath = '/glade/p/acom/acom-climate/cesm2/inputdata/atm/cam/ozone_strataero'

prescribed_strataero_file = 'ozone_strataero_WACCM_L70_zm5day_18500101-20150103_noPinatubo_c20220807.nc'

prescribed_ozone_datapath = '/glade/p/acom/acom-climate/cesm2/inputdata/atm/cam/ozone_strataero'

prescribed_ozone_file = 'ozone_strataero_WACCM_L70_zm5day_18500101-20150103_noPinatubo_c20220807.nc'

~Nan Rosenbloom

nanr@ucar.edu

ORCID ID =0000-0001-7389-3346

She/her/hers (Why do I share this info?)

[Quoted text hidden]

Gokhan Danabasoglu <gokhan@ucar.edu>

Wed, Mar 8, 2023 at 7:38 AM

To: Nan Rosenbloom <nanr@ucar.edu>

Cc: Mike Mills <mmills@ucar.edu>, Isla Simpson <islas@ucar.edu>, Gokhan Danabasoglu <gokhan@ucar.edu>

Nan,

Could you please let me know what the status is with this effort?

Best,
Gokhan
[Quoted text hidden]

Nan Rosenbloom <nanr@ucar.edu>
To: Gokhan Danabasoglu <gokhan@ucar.edu>
Cc: Mike Mills <mmills@ucar.edu>, Isla Simpson <islas@ucar.edu>, Gokhan Danabasoglu <gokhan@ucar.edu>

Wed, Mar 8, 2023 at 9:25 AM

Hi Gokhan ~

I *think* we need guidance from Mike on what we need to adjust for a noKrakatoa simulation.
Mike may need to create new forcing files, analogous to the noPinatubo forcing files?

Nan

~Nan Rosenbloom
nanr@ucar.edu
ORCID ID =0000-0001-7389-3346
She/her/hers ([Why do I share this info?](#))

[Quoted text hidden]

Mike Mills <mmills@ucar.edu>
To: Nan Rosenbloom <nanr@ucar.edu>
Cc: Gokhan Danabasoglu <gokhan@ucar.edu>, Isla Simpson <islas@ucar.edu>

Wed, Mar 8, 2023 at 10:24 AM

Sure. I'm looking into it now.

Mike

Mike Mills ([he/him/his](#))
CESM WACCM Community Liaison
Atmospheric Chemistry Observations & Modeling Laboratory
National Center for Atmospheric Research
Phone: 303-497-1425
<https://acomstaff.acom.ucar.edu/mmills/>
CESM forum: <https://bb.cgd.ucar.edu/cesm/>
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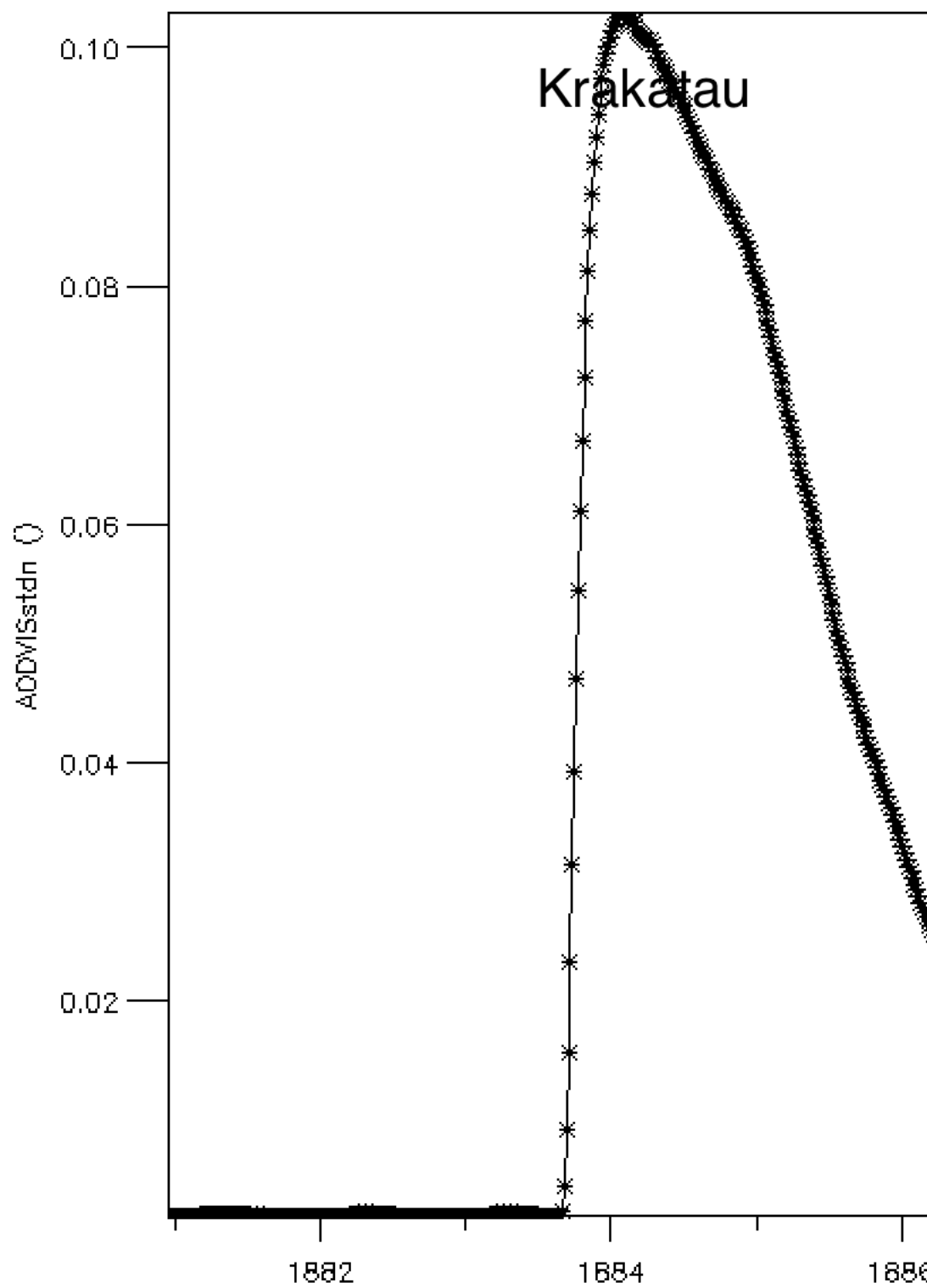
Mike Mills <mmills@ucar.edu>
To: Nan Rosenbloom <nanr@ucar.edu>, Gokhan Danabasoglu <gokhan@ucar.edu>, Isla Simpson <islas@ucar.edu>

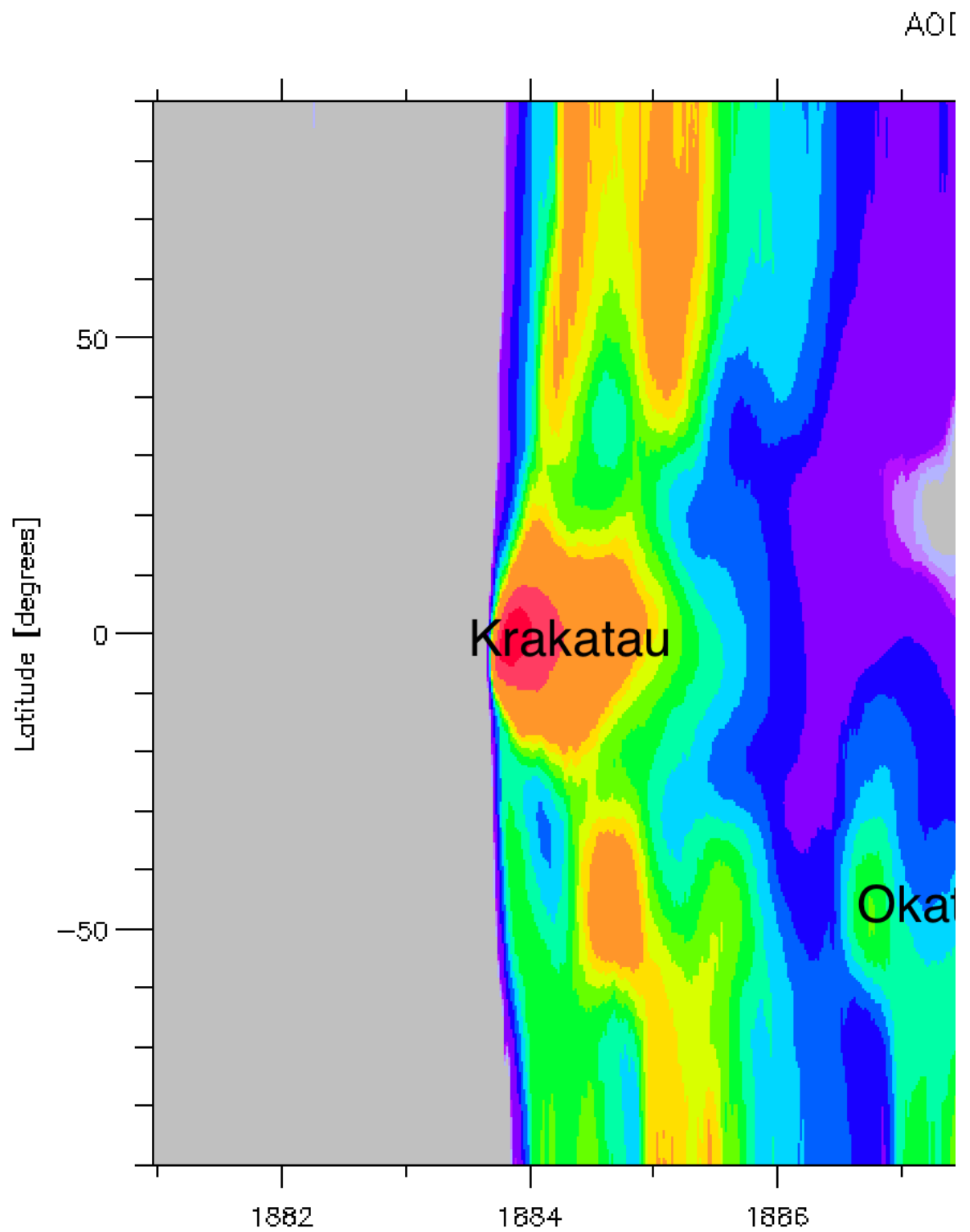
Okay, so I had a look at the forcing file I created for the "no Pinatubo" runs, as well as our emails from last August. It turns out that I actually did re-run WACCM without Pinatubo for 8 years (1: the Pinatubo eruption.

In the case of Krakatau (1883-08-27, 6°S, 9.4 Tg SO₂), the next eruption is 3 years later (Okataina, 1886-06-10, 38°S, 1.2 Tg SO₂), but this is still in the tail of decay from Krakatau. The most
Otherwise, if we want to patch in the piControl to erase Krakatau, it is going to be difficult to leave Okataina in without having some residue from Krakatau (option 2).

Option 3 is to erase both Krakatau and Okataina. It depends what your objectives are.

I am happy to run WACCM again to do option 1, or patch in the piControl for either option 2 or 3. Let me know what you think, given this further information.





/global/p/ocom/ocom-climate/cesm2/inputdata/atm/com/oxone_stratoera/oxone_stratoera_WACCM_L70_lm5day_18500101-20

*Mike***Mike Mills** (he/him/his)**CESM WACCM Community Liaison**

Atmospheric Chemistry Observations & Modeling Laboratory

National Center for Atmospheric Research

Phone: 303-497-1425

<https://acomstaff.acom.ucar.edu/mmills/>**CESM forum:** <https://bb.cgd.ucar.edu/cesm/>

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Gokhan Danabasoglu <gokhan@ucar.edu>

Wed, Mar 8, 2023 at 1:02 PM

To: Mike Mills <mmills@ucar.edu>

Cc: Nan Rosenbloom <nanr@ucar.edu>, Isla Simpson <islas@ucar.edu>, Gokhan Danabasoglu <gokhan@ucar.edu>

Mike,

Thank you. Based on your additional information, I am now favoring option #1. This follows the no-Pinatubo approach with the caveat that there is only one ensemble member. Isla?

Best,
Gokhan

[Quoted text hidden]

Isla Simpson <islas@ucar.edu>

Wed, Mar 8, 2023

To: Gokhan Danabasoglu <gokhan@ucar.edu>

Cc: Mike Mills <mmills@ucar.edu>, Nan Rosenbloom <nanr@ucar.edu>, Clara Deser <cdeser@ucar.edu>, Who Kim <whokim@ucar.edu>

Thanks for looking into it. At some point you asked Who and Clara (now cc'd) whether we should eliminate that other eruption too. Who's response is pasted in below, where as can also see different AOD graphic from those in Mike's email.

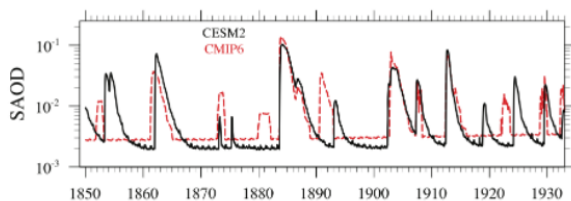
My feeling is that we should remove both Krakatau and Okataina motivated by the fact that we would then minimize any potential interaction between volcanic forcing and the ocean state in the period, unless there are compelling arguments for fully isolating Krakatau's influence from Okataina at this point. I think we could worry about separating out Krakatau specifically if we felt the need after demonstrating with as big a perturbation as possible, that it matters by removing both volcanoes.

But it would be good to hear what Clara and Who think given that they have spend much more time looking at the data.

Thanks,
Isla

I didn't know there was another eruption just after Krakatoa. Now it makes sense why the AOD signal from the Krakatoa eruption is more sustained than other eruptions (now I can see a smaller secondary peak!; see below SAOD plot from you). Because this plot suggests the influence of both eruptions overlaps, I wonder if each forcing is clearly separated from another...

Who



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Gokhan Danabasoglu <gokhan@ucar.edu>

Wed, Mar 8, 2023 at 2:01 PM

To: Isla Simpson <islas@ucar.edu>

Cc: Mike Mills <mmills@ucar.edu>, Nan Rosenbloom <nanr@ucar.edu>, Clara Deser <cdeser@ucar.edu>, Who Kim <whokim@ucar.edu>, Gokhan Danabasoglu <gokhan@ucar.edu>

Isla and All,

Just a correction..... I misread Mike's option #1. I was advocating using the same approach as used in the no-Pinatubo forcing construction. In this case, as both Clara and Who already indicated, we would remove both of the volcanoes.

Best,
Gokhan

[Quoted text hidden]

Mike Mills <mmills@ucar.edu>

Wed, Mar 8, 2023 at 2:07 PM

To: Gokhan Danabasoglu <gokhan@ucar.edu>

Cc: Isla Simpson <islas@ucar.edu>, Nan Rosenbloom <nanr@ucar.edu>, Clara Deser <cdeser@ucar.edu>, Who Kim <whokim@ucar.edu>

To clarify, for "no Pinatubo" I ran WACCM again without Pinatubo, but with Cerro Hudson. So that is consistent with option 1.

If you want to remove Okataina, I don't think we need to run WACCM again. We can paste in a non-volcanic period.

I also am realizing that the piControl has averaged volcanoes, so we would not paste that in. I could find a volcanically quiescent period from after 1850.

*Mike***Mike Mills, CESM WACCM Community Liaison**

Atmospheric Chemistry Observations & Modeling Laboratory

National Center for Atmospheric Research

mobile phone: 303.882.7089

<https://acomstaff.acom.ucar.edu/mmills/>**CESM forum:** <https://xenforo.cgd.ucar.edu/cesm/>

On Mar 8, 2023, at 2:02 PM, Gokhan Danabasoglu <gokhan@ucar.edu> wrote:

Isla and All,

Just a correction..... I misread Mike's option #1. I was advocating using the same approach as used in the no-Pinatubo forcing construction. In this case, as both Clara and Who already indicated, we would remove both of the volcanoes.

Best,
Gokhan

On Wed, Mar 8, 2023 at 1:52 PM Isla Simpson <islas@ucar.edu> wrote:

Thanks for looking into it. At some point you asked Who and Clara (now cc'd) whether we should eliminate that other eruption too. Who's response is pasted in below, where as can also see a slightly different AOD graphic from those in Mike's email.

My feeling is that we should remove both Krakatau and Okataina motivated by the fact that we would then minimize any potential interaction between volcanic forcing and the ocean state in this time period, unless there are compelling arguments for fully isolating Krakatau's influence from Okataina at this point. I think we could worry about separating out Krakatau specifically if we felt there was a need after demonstrating with as big a perturbation as possible, that it matters by removing both volcanoes.

But it would be good to hear what Clara and Who think given that they have spend much more time looking at the data.

Thanks,
Isla

<Screenshot at 2023-03-08 13-43-44.png>

On Wed, Mar 8, 2023 at 1:03 PM Gokhan Danabasoglu <gokhan@ucar.edu> wrote:

Mike,

Thank you. Based on your additional information, I am now favoring option #1. This follows the no-Pinatubo approach with the caveat that there is only one ensemble member. Isla?

Best,
Gokhan

On Wed, Mar 8, 2023 at 10:49AM Mike Mills <mmills@ucar.edu> wrote:

Okay, so I had a look at the forcing file I created for the "no Pinatubo" runs, as well as our emails from last August. It turns out that I actually did re-run WACCM without Pinatubo for 8 years (1991-1998) and merged the output into the default file. This was particularly important because there was a significant eruption in Chile (Cerro Hudson, August 15, 1991) exactly 2 months after Pinatubo, and a number of other small eruptions in the period of decay from the Pinatubo eruption.

In the case of Krakatau (1883-08-27, 6°S, 9.4 Tg SO₂), the next eruption is 3 years later (Okataina, 1886-06-10, 38°S, 1.2 Tg SO₂), but this is still in the tail of decay from Krakatau. The most rigorous procedure would be to just re-run WACCM again without Krakatau (option 1).

Otherwise, if we want to patch in the piControl to erase Krakatau, it is going to be difficult to leave Okataina in without having some residue from Krakatau (option 2).

Option 3 is to erase both Krakatau and Okataina. It depends what your objectives are.

I am happy to run WACCM again to do option 1, or patch in the piControl for either option 2 or 3. Let me know what you think, given this further information.

<2023-03-08 at 10.29.19 AM.png>

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Gokhan Danabasoglu <gokhan@ucar.edu>

Wed, Mar 8, 2023 at 2:19 PM

To: Mike Mills <mmills@ucar.edu>

Cc: Isla Simpson <islas@ucar.edu>, Nan Rosenbloom <nanr@ucar.edu>, Clara Deser <cdeser@ucar.edu>, Who Kim <whokim@ucar.edu>, Gokhan Danabasoglu <gokhan@ucar.edu>

Mike,

Thank you. I think going with pasting in a quiet period for the period with both makes sense. Let's go with that unless there is an objection.

Best,
Gokhan
[Quoted text hidden]

Mike Mills <mmills@ucar.edu>

To: Gokhan Danabasoglu <gokhan@ucar.edu>

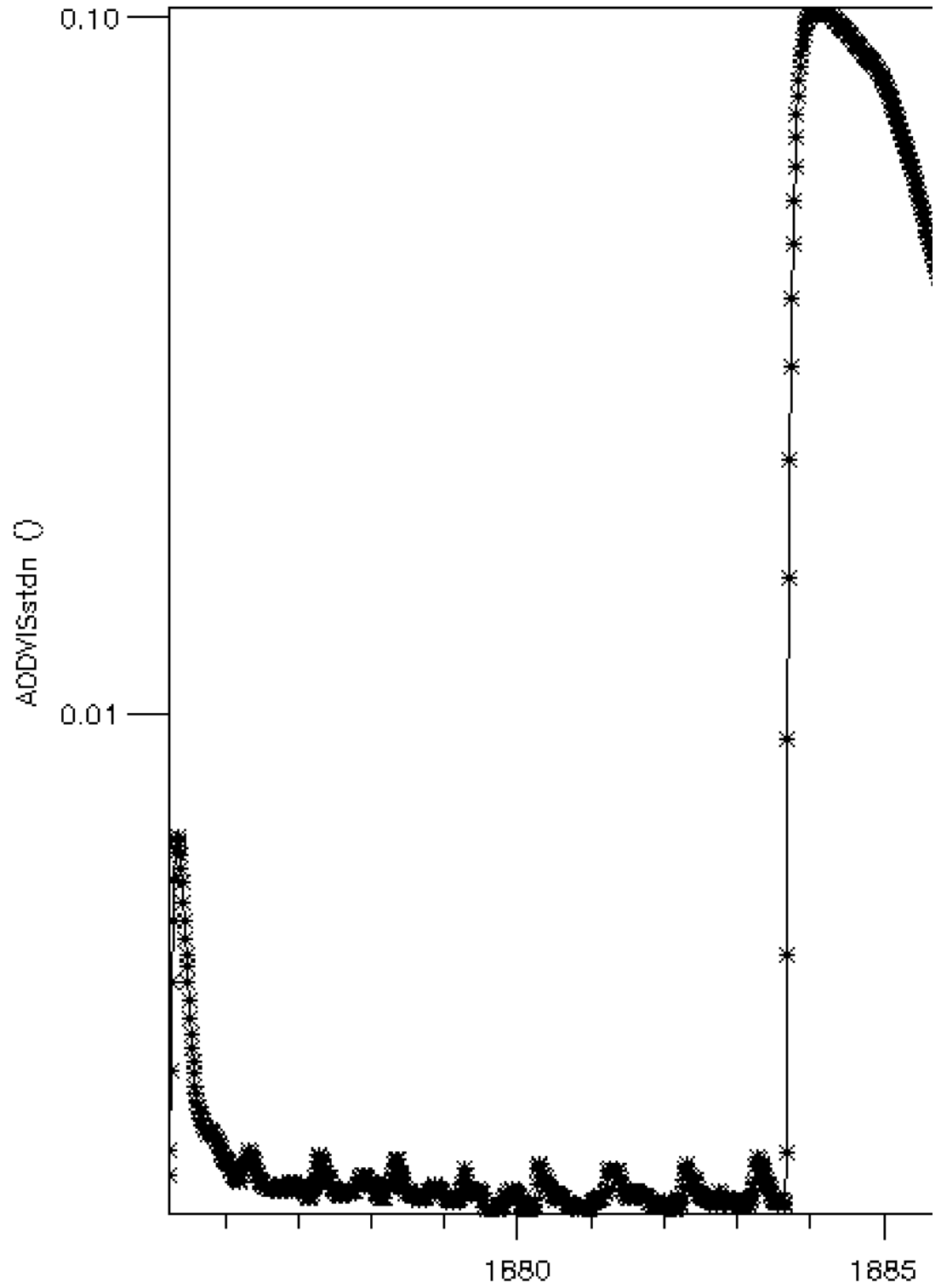
Cc: Isla Simpson <islas@ucar.edu>, Nan Rosenbloom <nanr@ucar.edu>, Clara Deser <cdeser@ucar.edu>, Who Kim <whokim@ucar.edu>

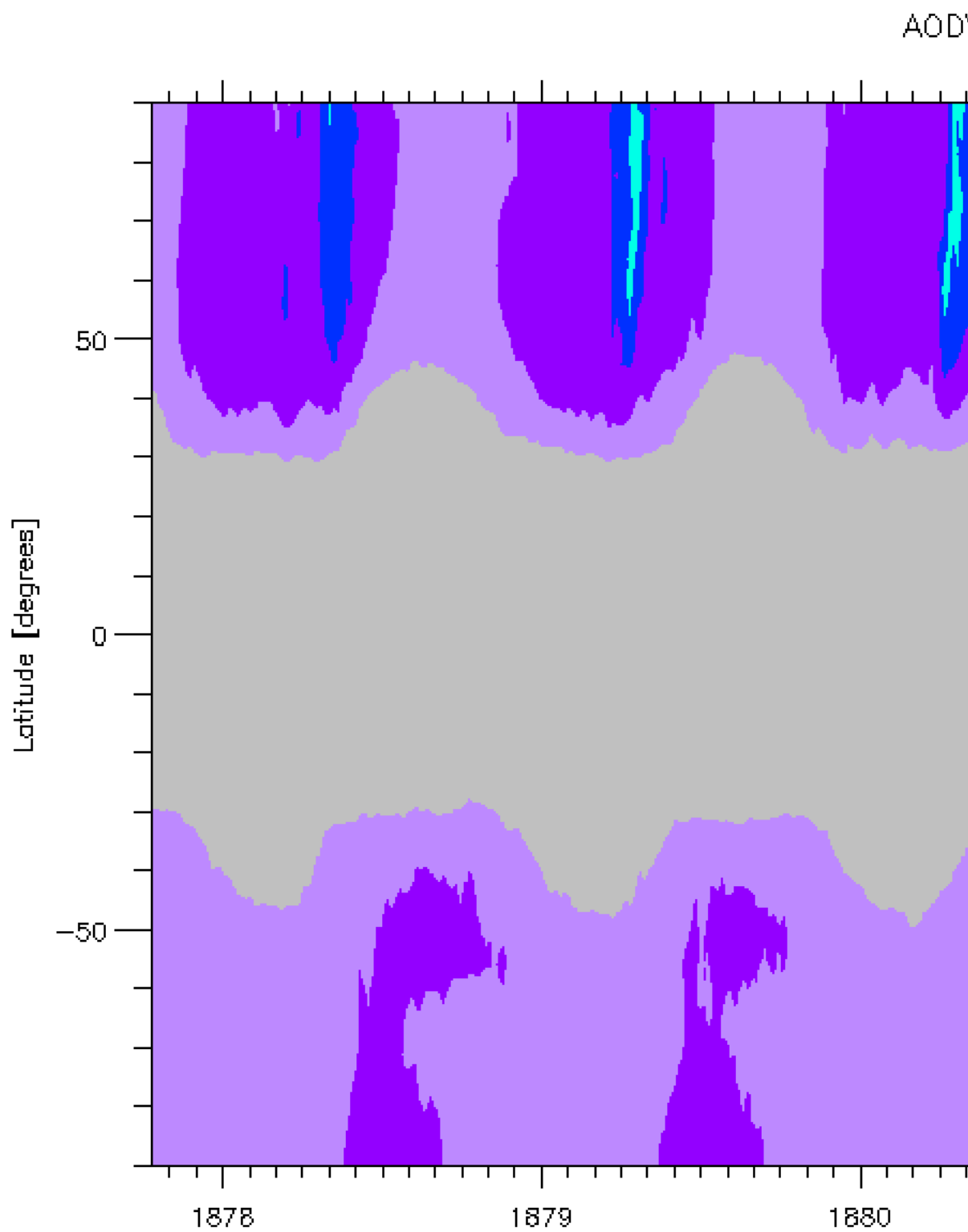
Sounds good. There is a 5-year quiet period right before Krakata (1878-1882). I will paste this forcing two times to fill the 10-year period 1883-1892.

I will start pasting with the first forcing date after Krakatau (which erupted August 27, 1883), matching with the corresponding date in 1878.

I will stop pasting at the end of 1882, since Calbuco erupted on January 7, 1893, and the forcing is 5-day averages.

Mike





/gloda/p/acorn/acorn-climate/acorn2/inputdata/atm/com/ozone_stratoera/ozone_stratoera_WACCM_L70_lm5day_18500101-2015

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Gokhan Danabasoglu <gokhan@ucar.edu>

Thu, Mar 9, 2023 at 7:06 AM

To: Mike Mills <mmills@ucar.edu>

Cc: Isla Simpson <islas@ucar.edu>, Nan Rosenbloom <nanr@ucar.edu>, Clara Deser <cdeser@ucar.edu>, Who Kim <whokim@ucar.edu>, Gokhan Danabasoglu <gokhan@ucar.edu>

Thank you very much, Mike.

Best,
Gokhan

[Quoted text hidden]

Mike Mills <mmills@ucar.edu>

To: Gokhan Danabasoglu <gokhan@ucar.edu>, Nan Rosenbloom <nanr@ucar.edu>, Isla Simpson <islas@ucar.edu>

Cc: Clara Deser <cdeser@ucar.edu>, Who Kim <whokim@ucar.edu>

The "no Krakatau" forcing file is created, checked, and ready to use.

user_nl_cam:prescribed_strataero_datapath = '/glade/p/acorn/acorn-climate/cesm2/inputdata/atm/cam/ozone_strataero'
prescribed_strataero_file = 'ozone_strataero_WACCM_L70_zm5day_18500101-20150103_noKrakatau1883_c20230309.nc'prescribed_ozone_datapath = '/glade/p/acorn/acorn-climate/cesm2/inputdata/atm/cam/ozone_strataero'
prescribed_ozone_file = 'ozone_strataero_WACCM_L70_zm5day_18500101-20150103_noKrakatau1883_c20230309.nc'

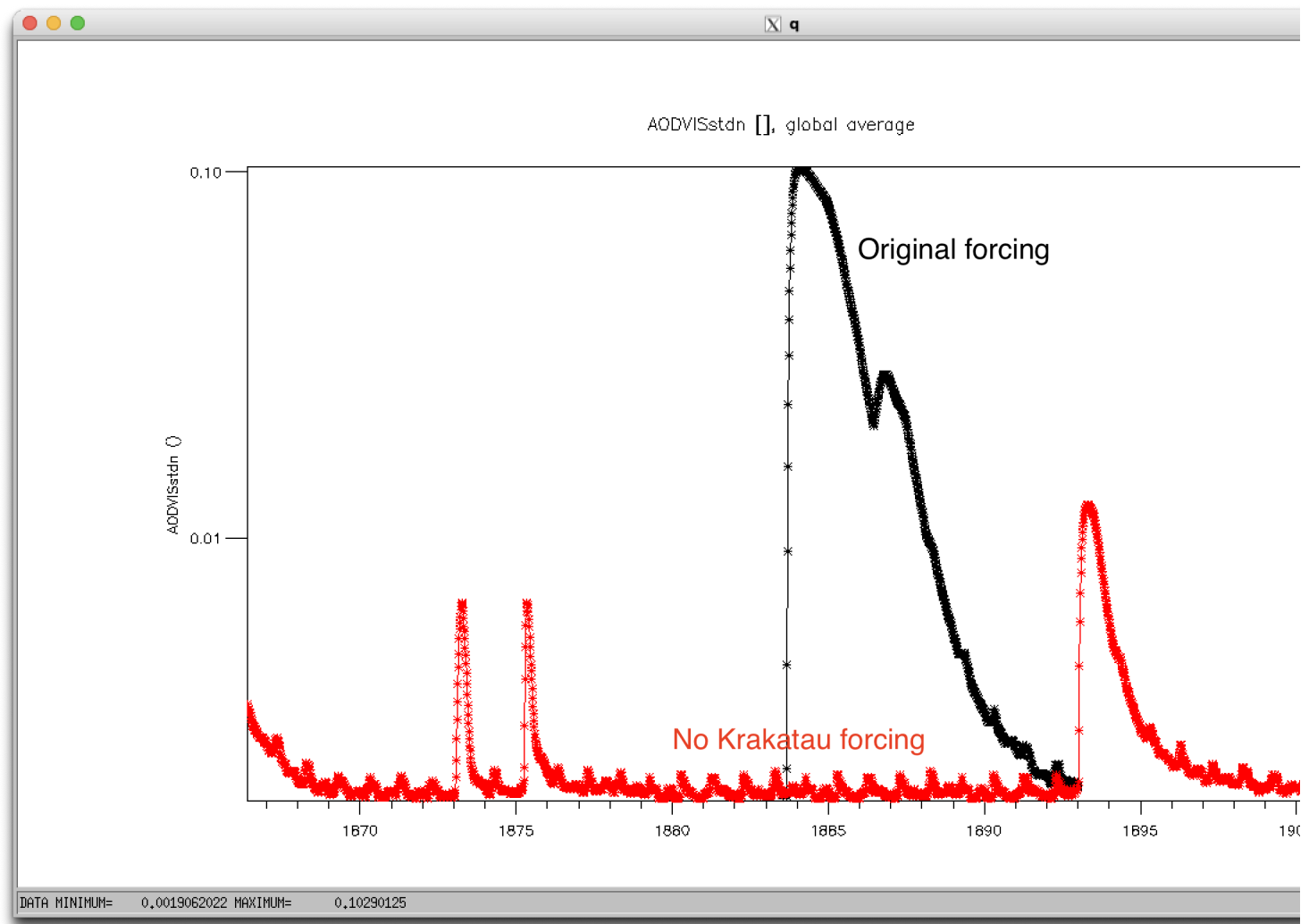
data_summary = "Removed eruptions Krakatau (1883-08-27) and Okataina (1886-06-10) from original forcing file\n",

"Merged data from dates 18830717=18830826 with data 5 years previous\n",

"Replaced data from 18830831-18880826 with 18780831-18830826\n",

"Replaced data from 18880831-18921119 with 18780831-18821119\n",

"Merged data from dates 18921124=18930103 with dates 10 years previous";

*Mike***Mike Mills** (he/him/his)

CESM WACCM Community Liaison

Atmospheric Chemistry Observations & Modeling Laboratory

National Center for Atmospheric Research

Phone: 303-497-1425

<https://acomstaff.ucar.edu/mmills/>CESM forum: <https://bb.cgd.ucar.edu/cesm/>

On Mar 9, 2023, at 7:06 AM, Gokhan Danabasoglu <gokhan@ucar.edu> wrote:

Thank you very much, Mike.

Best,
Gokhan

On Wed, Mar 8, 2023 at 3:43 PM Mike Mills <mmills@ucar.edu> wrote:

Sounds good. There is a 5-year quiet period right before Krakata (1878-1882). I will paste this forcing two times to fill the 10-year period 1883-1892.

I will start pasting with the first forcing date after Krakatau (which erupted August 27, 1883), matching with the corresponding date in 1878.

I will stop pasting at the end of 1882, since Calbuco erupted on January 7, 1893, and the forcing is 5-day averages.

Mike

<2023-03-08 at 3.33.16 PM.png>
<2023-03-08 at 3.31.01 PM.png>

[Quoted text hidden]

Nan Rosenbloom <nanr@ucar.edu>Thu, Mar 9, 2023 at 4:33 PM

To: Mike Mills <mmills@ucar.edu>
Cc: Gokhan Danabasoglu <gokhan@ucar.edu>, Isla Simpson <islas@ucar.edu>, Clara Deser <cdeser@ucar.edu>, Who Kim <whokim@ucar.edu>

Great! Thank you Mike.

Setup Question:
Gokhan: Does it matter which simulations we use to initialize the runs.

Presumably, these will be very similar to the no-Pinatubo simulations. The idea would be to start from the nearest restart files before the eruption and run them for 10-15 years or so. Ideally, we should do this for all four micro start years and 10 ensembles each.

We have these 1880 restarts on CS. Alternatively, we can ask for additional restarts from Busan:

drwxr-sr-x+ 3 strandwg cesm 4096 Apr 28 2022 b.e21.BHISTcmip6.f09_g17.LE2-1001.001/rest/1880-01-01-00000/
drwxr-sr-x+ 3 strandwg cesm 4096 Apr 27 2022 b.e21.BHISTcmip6.f09_g17.LE2-1021.002/rest/1880-01-01-00000/
drwxr-sr-x+ 3 strandwg cesm 4096 Apr 27 2022 b.e21.BHISTcmip6.f09_g17.LE2-1041.003/rest/1880-01-01-00000/
drwxr-sr-x+ 3 strandwg cesm 4096 Apr 27 2022 b.e21.BHISTcmip6.f09_g17.LE2-1061.004/rest/1880-01-01-00000/
drwxr-sr-x+ 2 strandwg cesm 4096 Apr 28 2022 b.e21.BHISTcmip6.f09_g17.LE2-1081.005/rest/1880-01-01-00000/
drwxr-sr-x+ 2 strandwg cesm 4096 Apr 28 2022 b.e21.BHISTcmip6.f09_g17.LE2-1101.006/rest/1880-01-01-00000/
drwxr-sr-x+ 2 strandwg cesm 4096 Apr 28 2022 b.e21.BHISTcmip6.f09_g17.LE2-1121.007/rest/1880-01-01-00000/
drwxr-sr-x+ 2 strandwg cesm 4096 Apr 27 2022 b.e21.BHISTcmip6.f09_g17.LE2-1141.008/rest/1880-01-01-00000/
drwxr-sr-x+ 2 strandwg cesm 4096 Apr 28 2022 b.e21.BHISTcmip6.f09_g17.LE2-1161.009/rest/1880-01-01-00000/
drwxr-sr-x+ 2 strandwg cesm 4096 Apr 27 2022 b.e21.BHISTcmip6.f09_g17.LE2-1181.010/rest/1880-01-01-00000/

~Nan Rosenbloom
nanr@ucar.edu
ORCID ID =0000-0001-7389-3346
She/her/hers (Why do I share this info?)

[Quoted text hidden]

Isla Simpson <islas@ucar.edu>Thu, Mar 9, 2023 at 6:24 PM

To: Nan Rosenbloom <nanr@ucar.edu>
Cc: Mike Mills <mmills@ucar.edu>, Gokhan Danabasoglu <gokhan@ucar.edu>, Clara Deser <cdeser@ucar.edu>, Who Kim <whokim@ucar.edu>

Thanks for checking. It's very important what the starting conditions are. I think the plan is to start from the macro's i.e., 1231, 1251, 1281 and 1301 and using 10 members of each, starting from the year that's closest to but before the Krakatoa eruption i.e., the latest year prior to 1883. Do you think it will be possible to get these from Busan?

Isla
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Nan Rosenbloom <nanr@ucar.edu>Thu, Mar 9, 2023 at 8:57 PM

To: Isla Simpson <islas@ucar.edu>
Cc: Mike Mills <mmills@ucar.edu>, Gokhan Danabasoglu <gokhan@ucar.edu>, Clara Deser <cdeser@ucar.edu>, Who Kim <whokim@ucar.edu>

I'll find out what we have available.

Thanks Isla!

Sent from my iPhone

On Mar 9, 2023, at 6:24 PM, Isla Simpson <islas@ucar.edu> wrote:

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Gokhan Danabasoglu <gokhan@ucar.edu>Fri, Mar 10, 2023 at 6:49 AM

3/15/23, 10:37 AM

NCARIUCAR Mail - No Krakatoa Runs

To: Nan Rosenbloom <nanr@ucar.edu>

Cc: Isla Simpson <islas@ucar.edu>, Mike Mills <mmills@ucar.edu>, Clara Deser <cdeser@ucar.edu>, Who Kim <whokim@ucar.edu>, Gokhan Danabasoglu <gokhan@ucar.edu>

Mike: Thank you for preparing the datasets.

Nan: I concur with Isla.

Best,
Gokhan

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Nan Rosenbloom <nanr@ucar.edu>

Fri, Mar 10, 2023 at 9:21 AM

To: Gokhan Danabasoglu <gokhan@ucar.edu>

Cc: Isla Simpson <islas@ucar.edu>, Mike Mills <mmills@ucar.edu>, Clara Deser <cdeser@ucar.edu>, Who Kim <whokim@ucar.edu>, Gokhan Danabasoglu <gokhan@ucar.edu>

Thanks all.

I have requested the restarts from Busan. I think the closest restart year will be 1880.

Will we want to run the simulation forward to create a new restart closer to the eruption year?

Nan

~Nan Rosenbloom

nanr@ucar.edu

ORCID ID =0000-0001-7389-3346

She/her/hers ([Why do I share this info?](#))

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Gokhan Danabasoglu <gokhan@ucar.edu>

Fri, Mar 10, 2023 at 9:49 AM

To: Nan Rosenbloom <nanr@ucar.edu>

Cc: Isla Simpson <islas@ucar.edu>, Mike Mills <mmills@ucar.edu>, Clara Deser <cdeser@ucar.edu>, Who Kim <whokim@ucar.edu>, Gokhan Danabasoglu <gokhan@ucar.edu>

Nan,

We can just use the 1880 restarts and run until the end of 1892. Since this length of simulation may not fit one submission, you may end up getting a restart in the middle of this segment.

Best,
Gokhan

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Nan Rosenbloom <nanr@ucar.edu>

Fri, Mar 10, 2023 at 10:00 AM

To: Gokhan Danabasoglu <gokhan@ucar.edu>

Cc: Isla Simpson <islas@ucar.edu>, Mike Mills <mmills@ucar.edu>, Clara Deser <cdeser@ucar.edu>, Who Kim <whokim@ucar.edu>, Gokhan Danabasoglu <gokhan@ucar.edu>

Sounds good.

Nan

~Nan Rosenbloom

nanr@ucar.edu

ORCID ID =0000-0001-7389-3346

She/her/hers ([Why do I share this info?](#))

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